

(L#8)

Shift Register :-

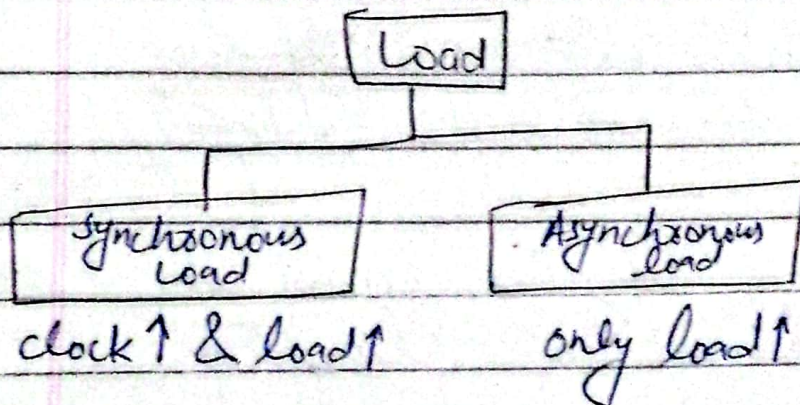
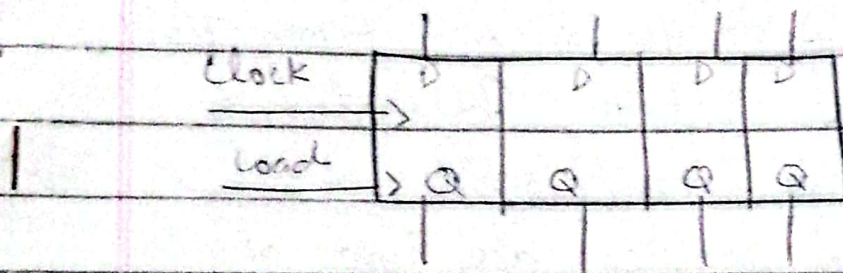
What is register?

Registers:-

→ Flip-flop is a 1-bit memory cell.

→ To increase the storage capacity, we have to use group of flip-flop. This group of flip-flop is known as register.

→ The n -bit register consist of ' n ' number of flip flop and is capable of storing ' n -bit' word.



Classification of Register

1) Depending on I/P & O/P

→ SISO

→ PISO

→ SIPO

→ PIPO

2) Depending on application

→ Shift registers

→ Storage registers

Shift Register

→ A register capable of shifting its binary information in one or both direction is called a shift register.

→ Shift registers are type of sequential circuit, mainly use for store or shift digital data.

→ Shift registers consist of an arrangement of flip-flop.

→ Shift register has no specified sequence of states.

Shift capacity:- The number of digital bits data, the shift register can retain (can be 1000) is called the shift capacity.
 $\text{no of stages} = \text{shift capacity}$

→ Typically, D-flip flop is used.

→ All flip-flops are driven by a common clock.

→ A shift register "shift" its stored data once every clock cycle.

→ All flip-flops are set or reset simultaneously.

Shift Register Operation

→ Shift left

→ Shift right

→ parallel load

→ No change / Hold

Serial-in-Serial out

$T.P = NT$

$N = \text{no of bits}$

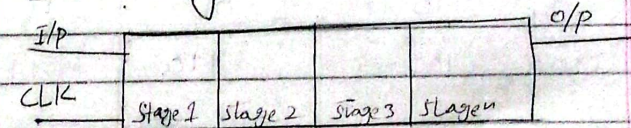
$T = \text{clock period}$

Shift register

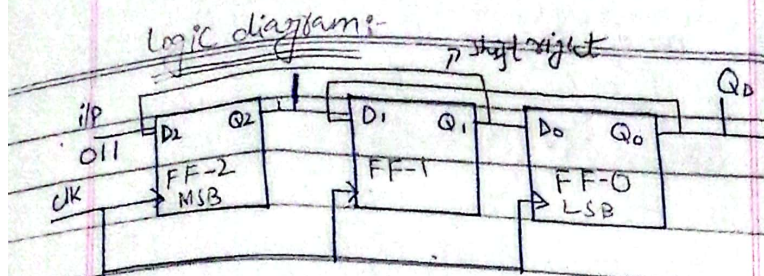
→ To store data n clocks are required.

→ To force data $n-1$ clocks are required.

Circuit diagram:- for total processing = $2n-1$ clock period



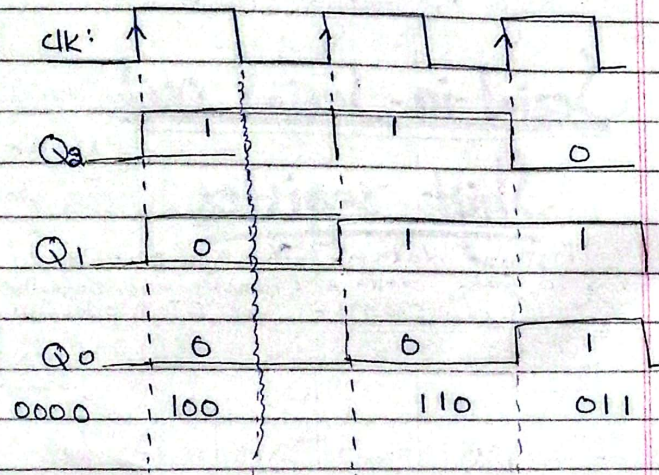
⇒ Used for long distance transmission.



Truth table:-

CLK	Q ₂	Q ₁	Q ₀
initially	0	0	0
↑	1	0	0
↑	1	1	0
↑	0	1	1

Wave form:-



Serial data VS parallel data movement

Serial	Parallel
→ Movement of N-bit data require N number of CLK pulses. Thus, the operation is slow. → Only one flip-flop is required to be connected at the output terminal, thus only one wire is required.	→ Require only one CLK to transfer all N-bit of data. Thus the operation is faster. → Required N number of connection to the output terminal, which is proportional to the number of bit. Thus too many connection is required.

Serial-in-Parallel out Register

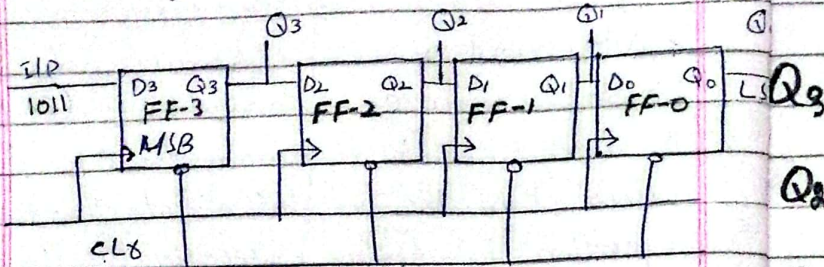
Logic Diagram:-

⇒ Disadvantage of SISO can be easily overcome by SIPO, where we don't need n-1 clock to take data

- n clock for storing data
- 0 clock for used of data
- n clock for total processing

It operates in two modes

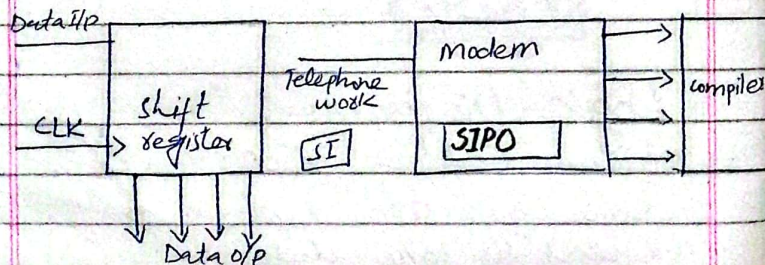
- Load mode
- Shift mode



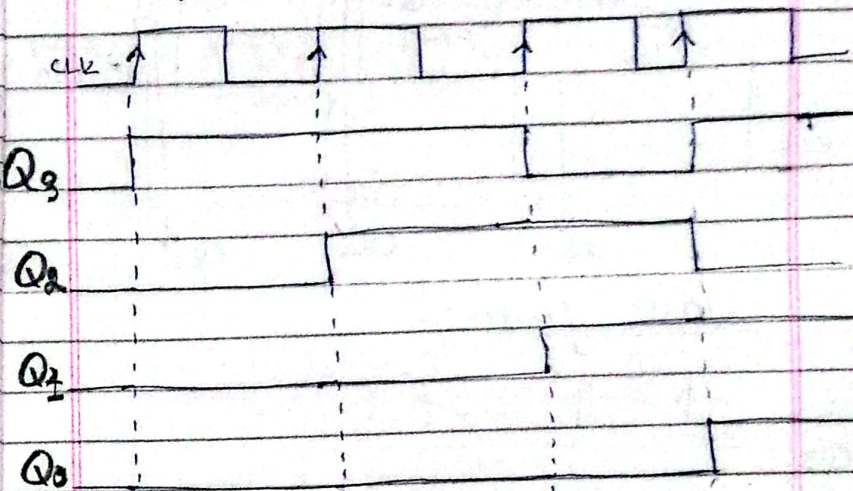
Truth table:-

CLK	I/P	Q3	Q2	Q1	Q0
initially	-	0	0	0	0
↑	1 → 1	0	0	0	0
↑	1 → 1	1	0	0	0
↑	0	0	1	1	0
↑	1	1	0	1	1

Application:-

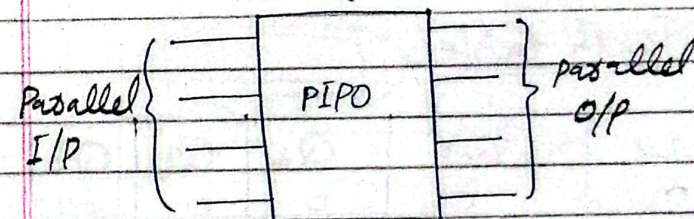


Wave form:-

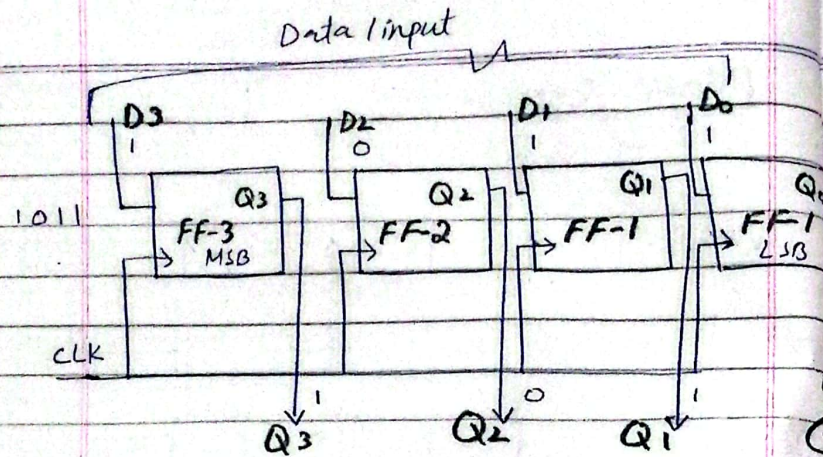


Parallel-in-Parallel out

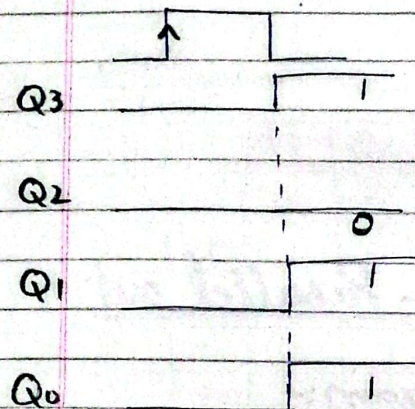
Block Diagram:-



→ It is storage registers or buffer registers.



Wave form:-

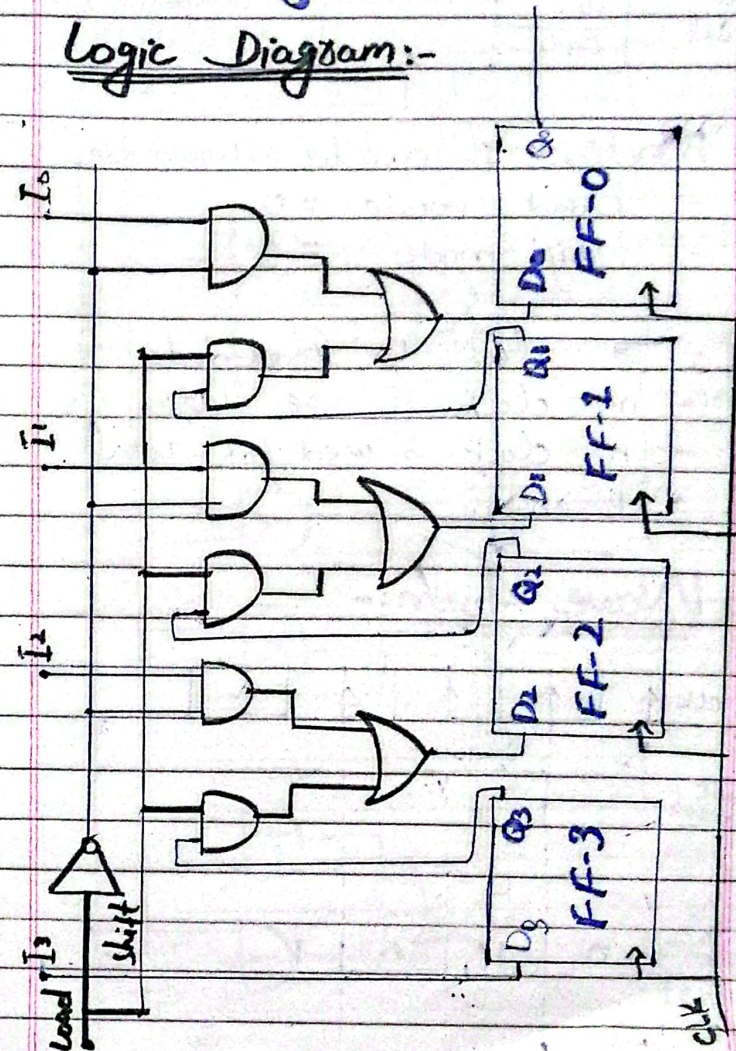


Truth table:

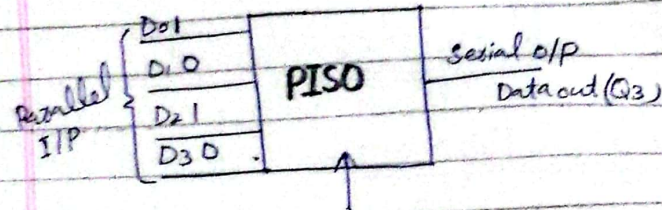
CLK	Q3	Q2	Q1	Q0
0	0	0	0	0
↑	1	0	1	1

Parallel-in-Serial out Registers

Logic Diagram:-



Block-Diagram:-

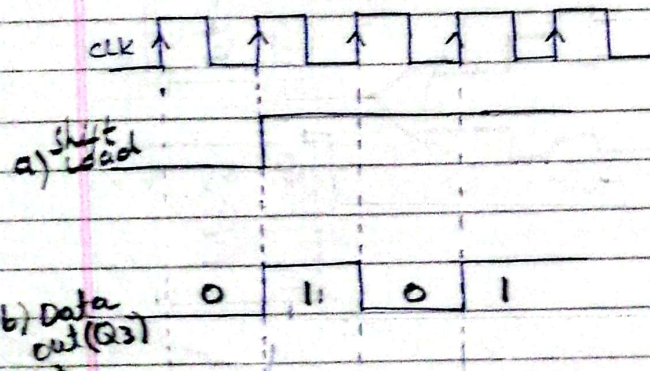


Modes:- It operates in two modes.

- Load mode = 0
- Shift mode = 1

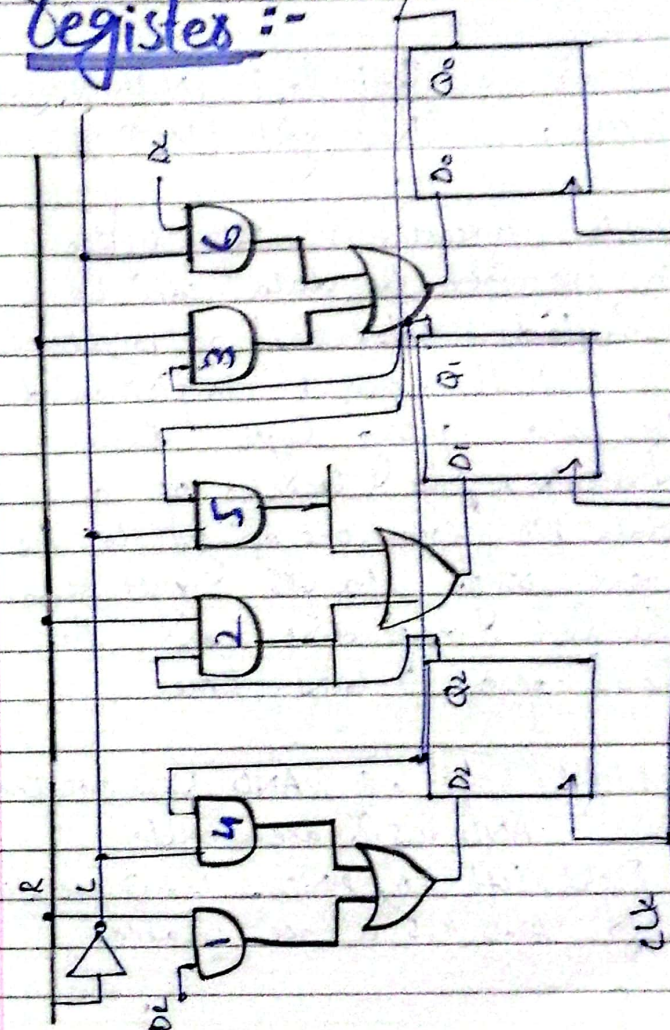
- 1 clock to load data.
- n-1 clock to use data.
- n clock is used for total processing.

Wave form:-

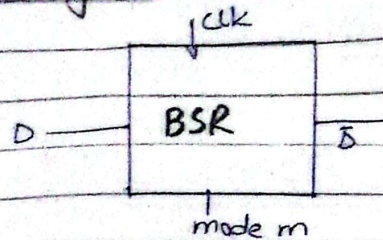


Truth table:-

Bidirectional Shift registers:-



logic diagram:-



$m=1$, Shift left (multiply by 2)
 $m=0$, Shift right (Divide by 2)

→ A bidirectional shift register is one in which the data can be shifted either left or right.

→ It can be implemented by using gating logic that enables the transfer of a data bit from one stage to the next stage to the right or to the left, depending on the level of a control line.

Right/Left = 1, AND 1, 2, 3 are Enable
 AND 4, 5, 6 are disable

Right/Left = 0, AND 1, 2, 3 are disable
 AND 4, 5, 6 are enable

Date 25/5/22

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Universal Shift Register

Definition:-

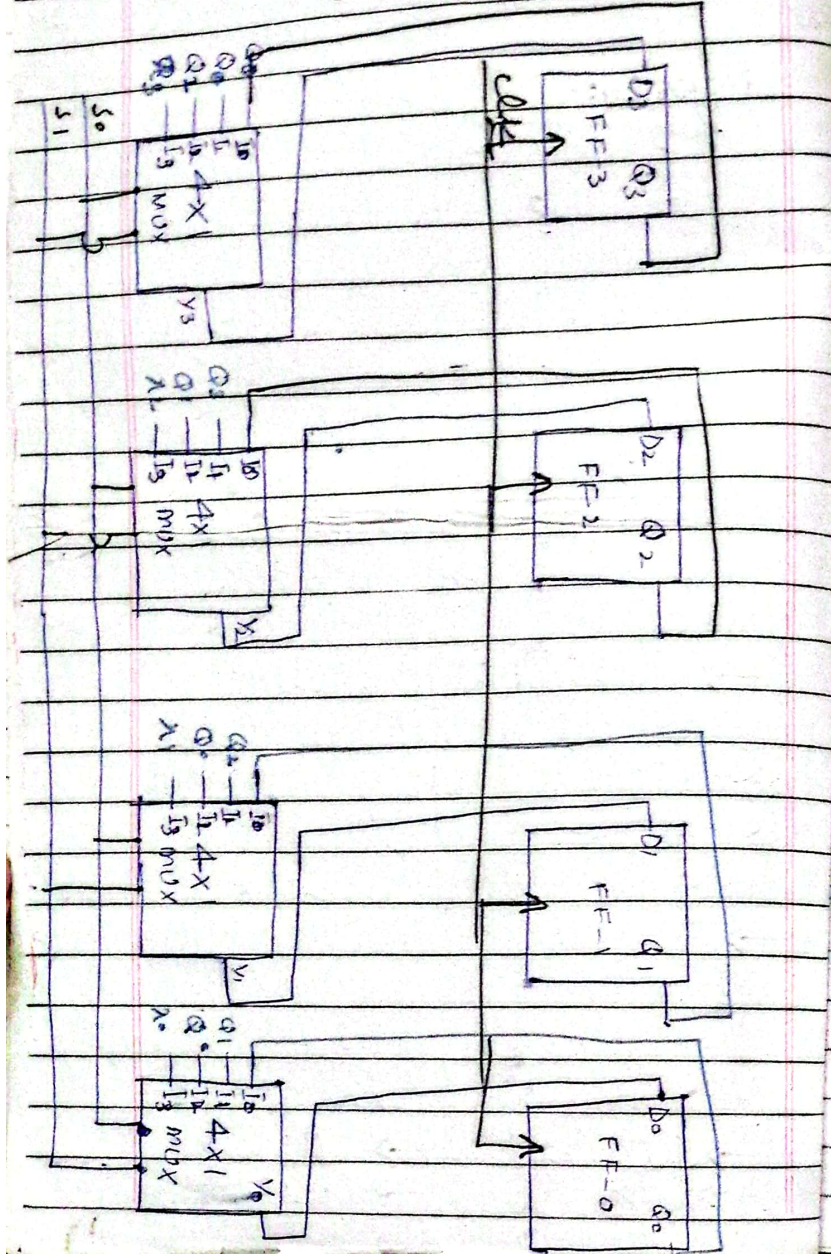
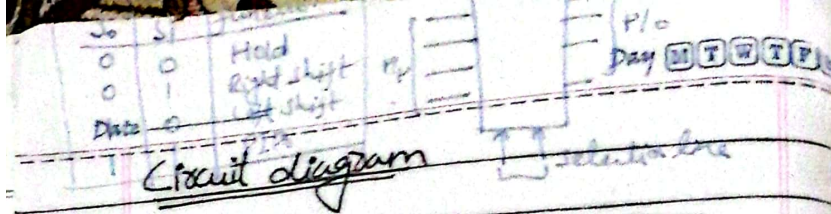
→ A register that can store the data and shifts the data towards the right and left along with parallel load capability is known as shift register.

→ It is used to perform input/output operations in both serial and parallel modes.

→ Unidirectional shift registers and bidirectional shift registers are combined together to get the design of the universal shift register.

→ It is also known as PIPO or shift register with the parallel output. It has four operations

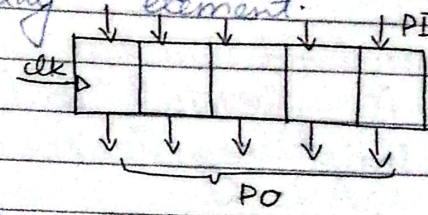
→ Shift left → Shift right
 → PIPO → Hold operation



#: What is PIPO Shift registers?

→ The shift register which allows parallel input (data is given separately to each flip flop and in simultaneous manner) and also produces a parallel output is known as PIPO shift register.

→ A PIPO shift register is used as temporary storage device and like SISO shift register it acts as delay element.

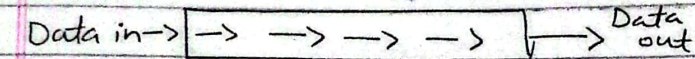


#: What is SISO shift registers?

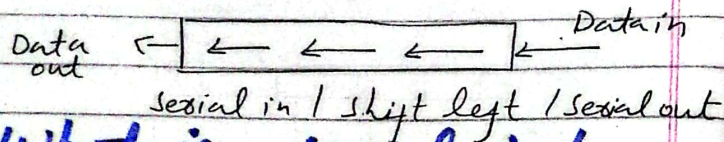
The shift register which allows serial input (one bit after the other through a single data line) and produces a serial output.

is called SISO shift register.
 → Since there is only one output, the data leaves the shift register one bit at a time in a serial pattern, thus the name SISO shift register.

The main use of SISO is act as a delay element.



i) Serial in / Shift right / Serial out

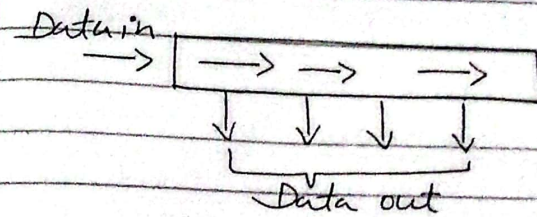


Q#1: What is serial in / PO shift register?

The shift register which allows serial input (one bit after the other through a single data line) and produces a parallel output is known as SIPO shift register.

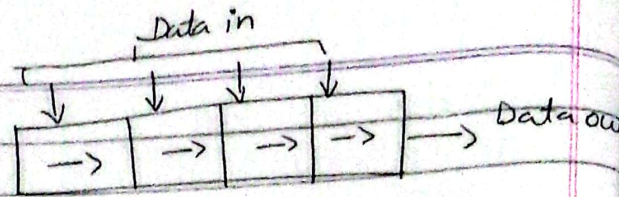
The output of first flip flop

is connected to the input of the next flip flop and so on.
 ⇒ They are used in communication lines where demultiplexing of a data line into several parallel lines is required because the main use of the SISO shift register is to convert serial input data into parallel data.



What is parallel In Serial out shift register?

⇒ A register which allows parallel input (data is given separately to each flip flop and in simultaneous manner) and produces a serial output is known as PISO shift register.



- PISO

Q# What is load terminal?
and types?

Load:-

To hold data in registers for long duration load terminal is used.

Synchronous load:-

In synchronous load, register is operational when load and clock are high.

Asynchronous load:-

In asynchronous load, register is operational when load is high.

Q# The group of bits 11001 is serially shifted into a 5-bit PO SR with an initial state 0110. After three clock pulses what will be the state?

LSB is Inverted and feedback to MSB

0110 → initial, 1011 → 1st CLK
0101 → second, 0010 → third CLK

Q# Assume that a 4-bit BISO shift register is initially clear. We wish to store the nibble 1100. What will be the 4-bit pattern after the second clock pulse?

In serial In serial out shift registers, data will be shifted at a time with every clock pulse. Therefore,

Wait / Store
1100 | 0000

110 | 0000 1st clock

11 | 0000 2nd clock

Q# A SIPO, 4-bit shift register initially contains all 1s. The data nibble is 0111 waiting to enter. After four clock pulses the register contains?

In SIPO shift register data will be shifted all at a time with every clock pulse. Therefore,

Wait	1	Store	
0111	1	0000	
011	1	1000	1st clock
01	1	1110	2nd clock
0	1	1110	3rd clock
X	1	1111	4th clock

Q# what will be the time if 200 kHz clock frequency, eight bits can be serially entered into a shift register in?

$$f = 200 \text{ kHz}$$

$$T = \frac{1}{f} = \frac{1}{200} \text{ msec} =$$

$$\left(\frac{1}{0.2}\right) (\text{micro-sec}) = \frac{200}{1} = 5 \text{ micro-sec}$$

In serial transmission, data enters one bit at a time.

After 8 clock cycles only 8 bit will be loaded
 $= 8 \times 5 = 40 \text{ usec}$

Q# what will be the time delay when 8-bit SISO shift register is used with a clock frequency of 2 MHz?

One clock period = $\frac{1}{2} \text{ usec} = 0.5 \text{ usec}$
 In serial transmission, data enters one bit at a time,
 So, the total delay = $0.5 \text{ usec} \times 8$
 $= 4 \text{ usec}$

is time required to transmit information of 8-bits.

On the fifth clock pulse, a 4-bit Johnson sequence is $Q_0=0, Q_1=1, Q_2=1$, and $Q_3=1$. On the sixth clock pulse, the sequence is -...?

$$Q_0=0, Q_1=0, Q_2=1, Q_3=1$$

Q# Difference b/w Serial and parallel digital data?

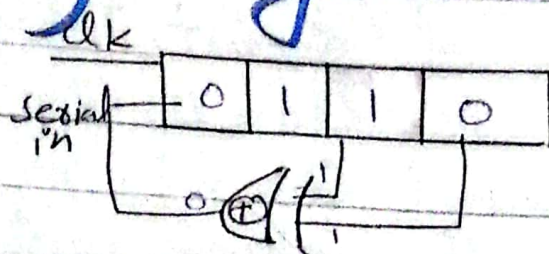
Serial:

Serial data is transmitted along one line, one bit at a time.

Parallel:

parallel data is transmitted all at once.

Q# The initial contents of the 4-bit SIPO shift register shown in figure is 0110 after three clock pulses are applied the contents of the shift register will be



CLK	inp	Q_4	Q_3	Q_2	Q_1
0	-	0	1	1	0
1	1 $\rightarrow$	1	0	1	1
2	0 $\rightarrow$	0	1	0	1
3	1 $\rightarrow$	1	0	1	0

Hence after 4 clock pulse the contents of register is 1010.

Q# How many clock pulses are required to enter a byte of data serially into an 8-bit shift register?

$\rightarrow$ 8-clock pulses are required.